MIL-S-43766A(GL)
26 September 1988
SUPERSEDING
MIL-S-43766(GL)
9 March 1972

MILITARY SPECIFICATION

SUPPORT ASSEMBLY, ARCH, INFLATABLE SHELTER SECTION (MUST)

This specification is approved for use within the U.S. Army Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of arch support assembly for the inflatable shelter section (MUST).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-P-1757 - Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5410

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PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-P-291 - Paperboard, Wrapping and Cushioning
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

MILITARY

MIL-T-704 - Treatment and Painting of Materiel
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-4-1609 - Support Assembly, Arch, Inflatable Shelter Section (MUST); Assembly Complete

5-4-1610 - Support Assembly, Arch Inflatable Shelter Section (MUST); Side, Center Arch and Cable Assemblies

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials. The materials used shall conform to the applicable referenced documents and requirements specified herein. Where materials are not definitely specified, they shall be of the quality and type normally used by the supplier provided the completed item complies with all provisions of this document.

* 3.2.1 Primer. Primer shall conform to type I or II, as applicable, color Y of TT-P-1757.

* 3.2.2 Enamel. Enamel shall conform to MIL-T-704 type F or G, as applicable. The color shall be No. 24554 of FED-STD-595.

3.3 Construction. Construction of the support assembly shall conform to Drawings 5-4-1609 and 5-4-1610 and as specified herein. All nuts shall be drawn up wrench tight, e.g. not being capable of being removed by hand.

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3.3.1 Welding. All welding shall be sound and smooth.

3.4 Finish.

3.4.1 Surface preparation. After forming and welding, aluminum arches, spacers, bolts, nuts, and washers shall be prepared for painting according to class 1A treatment of MIL-C-5541.

3.4.2 Priming. After surface preparation, the exterior surfaces of aluminum components shall be primed with the primer specified in 3.2.1. The inside surfaces of connectors and threaded surfaces of nuts and bolts shall be protected from receiving an excess of primer sufficient to prevent engagement or disengagement of mating component.

3.4.3 Enameling. After priming, the exterior surfaces of aluminum components shall be completely coated with the enamel specified in 3.2.2. The coating shall be smooth, uniform and dry film. The inside surfaces of connectors and threaded surfaces of nuts and bolts shall be protected from receiving an excess of enamel sufficient to prevent engagement or disengagement of mating components.

3.5 Marking for identification. Each support assembly shall be permanently and distinctly marked in the location as shown on Drawing 5-4-1609 with the manufacturer's identification or trademark of such known characters as to be identified with the manufacturer. The marking shall be not less than 1/2 inch in size and shall be legible after finishing.

3.6 Workmanship. The support assembly shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.2 In-process inspection. Inspection shall be made for the requirements listed in table I. Whenever nonconformance is noted, correction shall be made to the items affected, the lot in progress, and to the operation. Items that cannot be corrected shall be removed from production.

TABLE I. In-process inspection

Requirement or operation	Requirement paragraph
Welding is sound, smooth and is accomplished prior to finishing	3.3.1
Aluminum arches, spacers, bolts, nuts, and washers are prepared for painting	3.4.1
Aluminum components are primed prior to enameling	3.4.2

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4.4.3 End item visual examination. The end item shall be examined for the defects listed in table II. The lot size shall be expressed in units of support assemblies. The sample unit shall be one support assembly. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10.0 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not finished	101	
	Color or finish not as specified		201
	Coating not smooth or uniform		202
	Finish is not dry, i.e. wet or tacky to touch	102	
	Run, wrinkle, or blister		203
	Area of no film exposing bare metal	103	
	Area of thin film exposing primer coating		204
	Scratch, scuff, or foreign matter embedded in finish		205
	Inside surfaces of connectors or threaded surfaces of nuts and bolts have an excess of primer or enamel sufficient to prevent engagement or disengagement of mating components	104	
Construction and workmanship	Component missing	105	
	Component not fabricated as specified (e.g. end of "S" hook not closed around spacer)		206
	Nut not drawn up wrench tight (e.g. capable of being removed by hand)		207
	Component bent out of shape, is split, punctured, fractured, or twisted	106	
	Sharp edge or burr		208
Welding	Not sound and smooth	107	
	Burn through area, porous, or crack	108	
	Pits, slag inclusion, flash or excess flux deposit		209

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Marking for identification	Missing, illegible, less than 1/2 inch in size, or not located as shown on applicable drawing		210

4.4.4 End item dimensional examination. The end item shall be examined for the dimensions in the drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of support assemblies. The sample unit shall be one support assembly. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Bulged or distorted container The two side and one center arch assemblies not individually wrapped Scuffing or marring of the finish Cable assembly not neatly folded or rolled and secured to the side arch with twine or cloth tape Pad and bracket of the side arch assembly not covered with paper or other suitable wadding The two side and one center arch assemblies not compactly stacked, completely overwrapped and taped
Content	Number per shipping container is more or less than required

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. The two side and one center arch assemblies shall be individually wrapped in kraft paper to avoid scuffing or marring of the finish. The side arch with the permanently attached cable shall have the cable assembly neatly folded or rolled and secured to the side arch by means of twine or cloth tape. Each pad and bracket of the side arch assembly should be covered with paper or other suitable wadding for protection against damage to the finish. The two side and one center arch assemblies shall then be compactly stacked and completely overwrapped in single-faced corrugated fiberboard conforming to type F, style 1 of PPP-P-291 and taped with tape conforming to type III, class 2, grade B of PPP-T-45.

5.1.2 Commercial. Support assemblies shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Four support assemblies, preserved as specified in 5.1, shall be packed in a close-fitting fiberboard box conforming to style FTC, grade V2s of PPP-B-636. Each shipping container shall be closed, waterproofed, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636 except that inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four support assemblies, preserved as specified in 5.1, shall be packed in a close-fitting fiberboard box conforming to style FTC, variety DW, or SW, grade 275 of PPP-B-636. Each shipping container shall be closed and reinforced as specified in 5.2.1.

5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix PPP-B-636.

5.2.3 Commercial packing. Support assemblies, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

6.1 Intended use. The support assembly is intended for use with the inflatable shelter sections of Medical Unit, Self-contained, Transportable (MUST) to furnish support during a period when it is impossible to maintain internal pressure in the inflated tubes.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Subject term (key word) listing.

Arch support
Inflatable shelter
Support assembly

* 6.5 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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Preparing activity:

Army - GL

Project No. 5410-A323

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-43766A(GL)	Support Assembly, Arch, Inflatable Shelter Section (MUST)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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